MATHEMATICS

PROBABILITY
THEORY SHEET
ENGLISH

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1.1 IMPORTANT DEFINITIONS

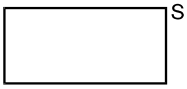
Random Experiment

It is a process which results in an outcome which is one of the various possible outcomes that are known to us before hand e.g. throwing of a die is a random experiment as it leads to fall of one of the outcome from {1, 2, 3, 4, 5, 6}. Similarly taking a card from a pack of 52 cards is also a random experiment.

Sample Space

It is the set of all possible outcomes of a random experiment e.g. {H, T} is the sample space associated with tossing of a coin.

In set notation it can be interpreted as the universal set.



Sample Point

Each element of the sample space is called a sample point.

Event

It is subset of sample space. e.g. getting a head in tossing a coin or getting a prime number in throwing a die.

Impossible Event

An event for which probability of its occurence is zero

Sure Event

An event for which probability of its occurence is 1

Equally likely Events

If events have same chance of occurrence, then they are said to be equally likely.

Exhaustive Events

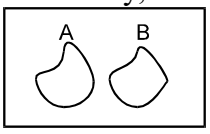
If each outcome of an experiment is associated with at least one of the events $E_1, E_2, E_3, ...E_n$, then collectively the events are said to be exhaustive. Mathematically we write

$E_1 \cup E_2 \cup E_3.....E_n = S$. (Sample space)

Mutually Exclusive Events

Two events are said to be mutually exclusive if occurrence of one of them rejects the possibility of occurrence of the other i.e. both cannot occur simultaneously.

In the vein diagram the events A and B are mutually exclusive. Mathematically, we write $A \cap B = \phi$



Events $A_1, A_2, A_3, ... A_n$ are said to be mutually exclusive events iff

$A_i \cap A_j = \phi \ \forall \ i, j \in \{1, 2, ..., n\}$ where $i \neq j$

Note : If $A_i \cap A_j = \phi \ \forall \ i, j \in \{1, 2, ..., n\}$ where $i \neq j$, then $A_1 \cap A_2 \cap A_3 .... \cap A_n = \phi$ but converse need not to be true.

Q.1 Write the sample space of the experiment 'A coin is tossed and a die is thrown'.

Ans.

Sol. The sample space S = {H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6}.

Q.2 Write the sample space of the experiment 'A coin is tossed, if it shows head a coin tossed again else a die is thrown.

Ans.

Sol. The sample space S = {HH, HT, T1, T2, T3, T4, T5, T6}

Q.3 Check which of the following events are equally likely, mutually exclusive or exhaustive

(i) If we throw a dice

Event A : Getting an even number; Event B : Getting an odd number

Sol. A = {2,4,6}

B = {1,3,5}

(a) Equally-likely	(b) Mutually exclusive	(c) Exhaustive
$P(A)=\frac{3}{6}$	$A \cap B = \phi$	$A \cup B = S$
$P(B)=\frac{3}{6}$	Yes	Yes

P(A) = P(B)

Yes

(ii) A ball is randomly drawn from an urn containing 2 white, 3 red and 4 green balls

Event A: Getting a white ball; Event B: Getting a red ball: Event C: Getting a green ball

Sol. $S = \{W_1, W_2, R_1, R_2, R_3, G_1, G_2, G_3, G_4\}$

$A = \{W_1, W_2\}$

$B = \{R_1, R_2, R_3\}$

$C = \{G_1, G_2, G_3, G_4\}$

(a) Equally likely	(b) Mutually Exclusive	(c) Exhaustive
$P(A)=\frac{2}{9}$	$A \cap B = \phi$	$A \cup B \cup C = S$
$P(B)=\frac{3}{9}$	$B \cap C = \phi$	Yes
$P(C)=\frac{4}{9}$	$C \cap A = \phi$	
No	Yes	

(iii) If we throw a dice

Event A : Getting an odd number; Event B : Getting a prime number

Sol. A = {1,3,5}

B= {2,3,5}

(a) Equally Likely	(b) Mutually Exclusive	(c) Exhaustive
$P(A)=\frac{3}{6}$	$A \cap B \neq \phi$	$A \cup B \neq S$
$P(B)=\frac{3}{6}$	No	No

Yes

1.2 DEFINITION OF PROBABILITY

We shall denote the number of out comes favourable to the event A by n(A) and the total number of out comes in the sample space S by n(S).

$\therefore \quad P(A) = \frac{n(A)}{n(S)}$.

Note:

All elements of sample space are equally likely, mutually exclusive and exhaustive

Complement of an Event

Complement of an event A is denoted by $\bar{A}$ or A' or A^C

$P(A^C) = 1 - P(A)$

Questions on definition of Probability

1. Four players P_1, P_2, P_3, P_4 are participating in a race. What is the probability that first two positions will be occupied by P_1 and P_2 respectively.

Ans. 1/12

Sol. Total = $\boxed{4}\boxed{3}\boxed{2}\boxed{1} = 24$

Favourable = $\overset{P_1}{\boxed{1}}\overset{P_2}{\boxed{1}}\boxed{2}\boxed{1} = 2$

$$P = \frac{2}{24} = \frac{1}{12}$$

2. From a pack of 52 cards, a card is randomly drawn. What is the probability that the card drawn is.

(i) Black card (ii) Diamond (iii) King (iv) Not a club

(v) Face card (vi) Diamond and a face card (vii) Diamond or a face card

(viii) Nither a diamond nor a King (ix) King or a red card

Ans. (i) 1/2 (ii) 1/4 (iii) 1/13 (iv) 3/4

(v) 3/13 (vi) 3/52 (vii) 22/52 (viii) 9/13

(ix) 7/13

3. Four apples and 3 oranges are arranged in a line. Find the probability that :

(i) All the oranges are together (ii) no two oranges are together

Ans. (i) 1/7

Sol. (i) Total = 4 - A & 3 - 0

$= 7!$

Favorable = $O_1O_2O_3, A_1, A_2, A_3, A_4$

$= 5! \times 3!$

$$P = \frac{5!3!}{7!} = \frac{1}{7}$$

(ii) Favourable = $__A_1__A_2__A_3__A_4__$

$= 4! \times {}^5C_3 \times 3!$

$$P = \frac{4!{}^5C_3 \times 3!}{7!} = \frac{2}{7}$$

4. Two natural numbers are randomly selected from the set of first 20 natural numbers what is the probability that
(i) both are even (ii) at least one is odd (iii) their sum is odd

Ans. (i) 9/38 (ii) 29/38 (iii) 10/19

Sol. (i) Total = $^{20}C_2$

Even = $^{10}C_2$

$$P = \frac{^{10}C_2}{^{20}C_2} = \frac{9}{38}$$

(ii) P(E) = 1 – P(when both are even)

$$= 1 - \frac{9}{38}$$

$$= \frac{29}{38}$$

(iii) One is even & one is odd

$$= \frac{^{10}C_1 \times ^{10}C_1}{^{20}C_2} = \frac{10}{19}$$

5. Seven different colours of a rainbow are arranged in a line. What is the probability that
(i) Green is always in the middle. (ii) Green, yellow and orange are always together

Ans. (i) 1/7 (ii) 1/7

Sol. (i) VIBGYOR

$$P = \frac{1 \times 6!}{7!} = \frac{1}{7}$$

(ii) GYO, V, I, B, R

$$= \frac{5! \times 3!}{7!} = \frac{1}{7}$$

6. Two numbers are randomly selected from set of first 6 natural numbers. What is the probability that selected pair is co-prime?

Ans. 11/15

Sol. 1,2,3,4,5,6

Total = $^6C_2 = 15$

Co-prime = 15 – 4{24,26,46,36}

= 11

$$P = \frac{11}{15}$$

7. A leap year is selected at random. Find the probability that it has
(i) 53 sundays (ii) 53 sundays and 53 mondays
(iii) 53 sundays or 53 mondays (iv) 53 sundays or 53 fridays
(v) 53 sundays and 53 fridays

Ans. (i) 2/7 (ii) 1/7 (iii) 3/7 (iv) 4/7
(v) 0

Sol. Leap year = 366 days

$\frac{366}{7}$ = 52 weeks & 2 days

For 2 days → Sun & mon
Mon & Tue,
Tue & Wed
Wed & Thurs.
Thur. & Fri
Fri & Sat
Sat & Sun.

Total 7 possibilities

1.3 ODDS

Odds in Favour and Odds Against an Event

As a result of an experiment, if p of the outcomes are favourable to an event E and q of the outcomes are against it, then we say that odds are p to q in favour of E or odds are q to p against E.

$\therefore$ Odds in favour of an event E = $\frac{\text{number of favourable cases}}{\text{number of unfavourable cases}} = \frac{p}{q}$

and odds against an event E = $\frac{\text{number of unfavourable cases}}{\text{number of favourable cases}} = \frac{q}{p}$

If p(E) = P

Odds in favour of an event E = $\frac{P}{1-P}$

and odds against an event E = $\frac{1-P}{P}$

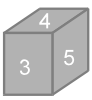
If odds in favour of an event E are p : q then the probability of the occurrence of that event is $\frac{p}{p+q}$

If odds against an event E are p : q the probability of the occurrence of that event is $\frac{q}{p+q}$

2. CREATION OF SAMPLE SPACE

Q.1 Find the sample space associated with the experiment of rolling a pair of dice (plural of die) once. Also find the number of elements of the sample space.





	1	2	3	4	5	6
1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
2	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
3	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
4	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
5	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
6	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

Number of elements (outcomes) of the above sample space is 6 × 6 i.e., 36

Find probability that

(i) Total is

(a) 2 (b) 3 (c) 4 (d) 5 (e) 6 (f) 7

(g) 8 (h) 9 (g) 10 (j) 11 (k) 12

Sol. (a) $\frac{1}{36}$ (b) $\Rightarrow P(E) = \frac{2}{36}$ (c) $\Rightarrow P(E) = \frac{3}{36}$ (d) $\Rightarrow P(E) = \frac{4}{36}$

(e) $\Rightarrow P(E) = \frac{5}{36}$ (f) $\Rightarrow P(E) = \frac{6}{36}$ (g) $\Rightarrow P(E) = \frac{5}{36}$

(h) $\Rightarrow P(E) = \frac{4}{36}$ (g) $\Rightarrow P(E) = \frac{3}{36}$ (j) $\Rightarrow P(E) = \frac{2}{36}$

(k) $\Rightarrow P(E) = \frac{1}{36}$

(ii) Total is atmost 4

Ans $P(E) = \frac{1+2+3}{36} = \frac{6}{36}$

(iii) total is atleast 7

Ans $P(E) = \frac{6+5+4+3+2+1}{36} = \frac{21}{36}$

(iv) total is atleast 5 & atmost 9

Ans $P(E) = \frac{4+5+6+5+4}{36} = \frac{25}{36}$

(v) total is prime

Ans $P(E) = \frac{1+2+4+6+2}{36} = \frac{15}{36}$

(vi) total is 6 & largest no is 4

Ans $P(E) = \frac{2}{36}$

Sol. Favourable = {24, 42}

$P = \frac{2}{36}$

(vii) total is 6 or largest no is 4

Ans $P(E) = \frac{10}{36}$

Sol. Favourable = {15,24,33,42,51,41,43,44,14,34}

$= \frac{10}{36}$

Q.2 Two dice are thrown and number appearing on them are different. What is the probability that :

(i) Total is 8 (ii) Number 3 appears at least once. (iii) There is no 5 (iv) Total exceeds 9

Ans. (i) 4/30 (ii) 10/30 (iii) 20/30 (iv) 4/30

Q.3 A coin is tossed 4 times, create sample space.

Find probability that

(i) These are exactly 2 heads (ii) These is no head

(iii) No. of heads > no of tails (iv) H & T accur alternatily

(v) No two consecutive heads (vi) H does not accur at first place

(vii) Length of longest string of H is 2 (viii) Length of longest string of H is 3

Ans. (i) $P(E)=\frac{6}{16}$ (ii) $P(E)=\frac{1}{16}$ (iii) $P(E)=\frac{5}{16}$ (iv) $P(E)=\frac{2}{16}$

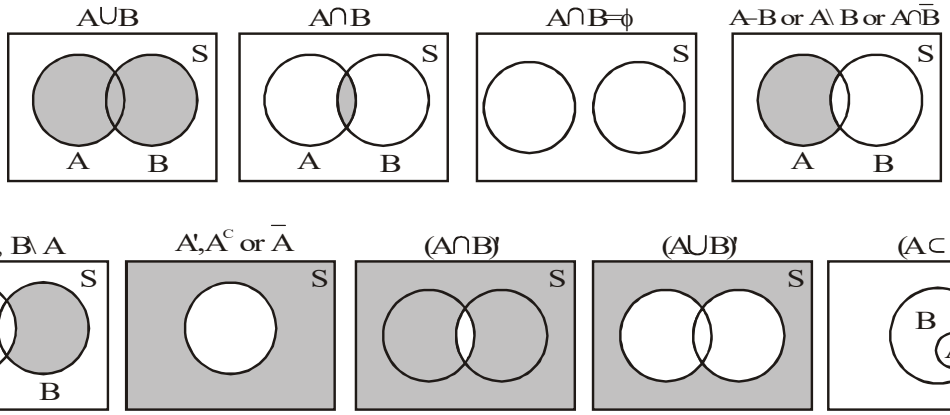
(v) $P(E)=\frac{8}{16}$ (vi) $P(E)=\frac{8}{16}$ (vii) $P(E)=\frac{5}{16}$ (viii) $P(E)=\frac{2}{16}$

Sol. $\{HHHH, HHHT, HHTH, HHTT, HTHH, HTHT, HTTH, HTTT, THHH, THHT, THTH, THTT, TTHH, TTHT, TTTH, TTTT\}$

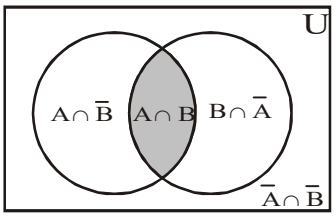
3. VENN DIAGRAM

A diagram used to illustrate relationships between sets. Commonly, a rectangle represents the universal set and a circle within it represents a given set (all members of the given set are represented by points within the circle). A subset is represented by a circle within a circle and intersection is indicated by overlapping circles.

Let S is the sample space of an experiment and A, B are two events corresponding to it :



ADDITION THEOREM :



$A \cup B = A + B$ = A or B denotes occurrence of at least A or B.

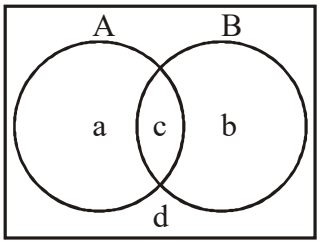
For 2 events A & B :

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

1. For any two events A and B. Let $P(A \cup B) = 7/8$; $P(A \cap B) = 1/4$ and $P(A^C) = 5/8$, Find P(A),P(B) and $P(A \cap B^c)$

Ans.

Sol.



$a + b + c + d = 1 \times 8 = 8$

$P(A \cup B) = \frac{7}{8} \Rightarrow a + b + c = 7$

$P(A \cap B) = \frac{1}{4} \Rightarrow c = 2$

$P(\bar{A}) = \frac{5}{8} \Rightarrow b + d = 5$

$\Rightarrow a = 1, b = 4, c = 2, d = 1$

$P(A) = \frac{a+c}{8} = \frac{3}{8}$

$P(B) = \frac{b+c}{8} = \frac{6}{8}$

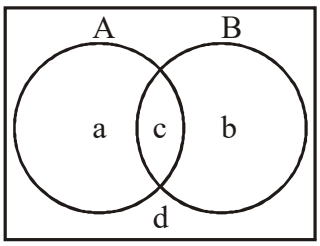
$P(A \cap \bar{B}) = \frac{a}{8} = \frac{1}{8}$

2. If A and B are any two events with $P(A) = 3/8$; $P(B) = 1/2$ and $P(A \cap B) = 1/4$. Find

(i) $P(A \cup B)$ (ii) $P(A^C)$ and $P(B^C)$ (iii) $P(A^C \cap B^C)$ (iv) $P(A^C \cup B^C)$ (v) $P(A \cap B^C)$ (vi) $P(B \cap A^C)$

Ans.

Sol $P(A) = \frac{3}{8}$ $P(B) = \frac{1}{2}$ $P(A \cap B) = \frac{1}{4}$



$a + b + c + d = 8$

$a + c = 3$

$b + c = 4$

$c = 2$

$a = 1, b = 2, c = 2, d = 3$

(i) $P(A \cup B) = \frac{1+2+2}{8} = \frac{5}{8}$

(ii) $P(\bar{A}) = \frac{5}{8}$ $P(\bar{B}) = \frac{1}{2}$

(iii) $P(\bar{A} \cap \bar{B}) = \frac{3}{8}$

(iv) $P(\bar{A} \cup \bar{B}) = \frac{a+b+d}{8} = \frac{3}{4}$

(v) $P(A \cap \bar{B}) = \frac{a}{8} = \frac{1}{8}$

(vi) $P(B \cap \bar{A}) = \frac{b}{8} = \frac{1}{4}$

3. If card is randomly drawn from a pack of 52 playing cards.

A = Red card is drawn B = Face card is drawn Find

(i) P(A) (ii) P(B) (iii) $P(A \cap B)$ (iv) $P(A \backslash B)$

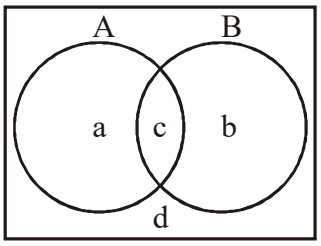
(v) $P(B \backslash A)$ (vi) $P\left(\bar{A} \cup \bar{B}\right)$

Ans. (i) 26/52 (ii) 12/52 (iii) 6/52 (iv) 20/52 (v) 6/52 (vi) 46/52

4. CONDITIONAL PROBABILITY

Let A and B be two events such that p (A) >0. Then p(B|A) denote the conditional probability of B given that A has occurred. Since A is known to have occurred, it becomes the new sample space replacing the original S. From this we led to the definition.

$p(B|A) = \frac{p(A \cap B)}{p(A)} = p(B/A) = p\left(\frac{B}{A}\right)$ which is called conditional probability of B given A



$P(A/B) = \frac{c}{b+c}$

$P(A/B) = \frac{P(A \cap B)}{P(B)}$

Eg : (1) $P(B/A) = \frac{c}{a+c}$ (2) $P(A/A \cup B) = \frac{a+c}{a+b+c}$ (3) $P(\bar{B}/\bar{A}) = \frac{d}{b+d}$

1. If card is randomly drawn from a pack of 52 playing cards.

A = Red card is drawn B = Face card is drawn Find

(i) P(A/B) (ii) P(B/A) (iii) $p(\bar{A}/\bar{B})$ (iv) $p(\bar{B}/A \cup B)$

Ans. (i) 1/2 (ii) 3/13 (iii) 1/2 (iv) 10/16

2. Given P(A) = 1/2 ; P(B) = 1/3 and $P(A \cap B) = 1/4$. Find

(i) P(A/B) (ii) P(B/A) (iii) $P(A \cup B)$

(iv) $P(A^C/B^C)$ (v) $P(B^C/A^C)$

Ans. (i) 3/4 (ii) 3/6 (iii) 7/12 (iv) 5/8 (v) 5/6

3. If we throw two dice

A = getting a doublet B = total is 10 or more

(i) P(A/B) (ii) P(B/A) (iii) $P(A/A \cup B)$ (iv) $P\left(A/\bar{B}\right)$

Ans. (i) 2/6 (ii) 2/6 (iii) 6/10 (iv) 4/30

4. Find P(B/A) if (i) A is subset of B (ii) A and B are disjoint

Ans. (i) 1 (ii) 0

5. MULTIPLICATION THEOREM

$P(A \cap B)=P(A)P(B/A)$ which is called compound probability or multiplication theorem. It says the probabilitythat both A and B occur is equal to the probability that A occur times the probability that B occurs given that Ahas occurred.

For any three events A_1, A_2, A_3 we have $P(A_1 \cap A_2 \cap A_3) = P(A_1)\, P(A_2|A_1)\, P(A_3|\,(A_1 \cap A_2))$

1. A bag contains 3 white and 6 red balls. Two balls are drawn one by one without replacement. Find the probability that

(i) both are red (ii) both are white

(iii) first is white and second is red (iv) one is white and one is red

Ans. (i) $\frac{6}{9}\times\frac{5}{8}$ or $\frac{{}^6C_2}{{}^9C_2}$ (ii) $\frac{3}{9}\times\frac{2}{8}$ or $\frac{{}^3C_2}{{}^9C_2}$ (iii) $\frac{3}{9}\times\frac{6}{8}$ (iv) $\left(\frac{3}{9}\times\frac{6}{8}\right)(2!)$ or $\frac{{}^3C_1\,{}^6C_1}{{}^9C_2}$

2. A bag contains 3 white and 6 red balls. Two balls are drawn one by one with replacement. Find the probability that

(i) both are red (ii) both are white

(iii) first is white and second is red (iv) one is white and one is red

Ans. (i) $\left(\frac{6}{9}\right)^2$ (ii) $\left(\frac{3}{9}\right)^2$ (iii) $\frac{3}{9}\times\frac{6}{9}$ (iv) $\left(\frac{3}{9}\times\frac{6}{9}\right)(2!)$

3. From pack of 52 cards 3 cards are drawn one by one without replacement. Find the probability that :

(i) all three are diamonds (ii) all three are king

(iii) first is king, second is queen and third is jack (iv) one is king, one is queen and one is jack

(v) first two are kings and third is queen (vi) there are two kings and one queen

Ans. (i) $\frac{13}{52}\times\frac{12}{51}\times\frac{11}{50}$ or $\frac{{}^{13}C_3}{{}^{52}C_3}$ (ii) $\frac{4}{52}\times\frac{3}{51}\times\frac{2}{50}$ or $\frac{{}^4C_3}{{}^{52}C_3}$ (iii) $\frac{4}{52}\times\frac{4}{51}\times\frac{4}{50}$

(iv) $\left(\frac{4}{52}\times\frac{4}{51}\times\frac{4}{50}\right)(3!)$ or $\frac{{}^4C_1\,{}^4C_1\,{}^4C_1}{{}^{52}C_3}$ (v) $\frac{4}{52}\times\frac{3}{51}\times\frac{4}{50}$ (vi) $\left(\frac{4}{52}\times\frac{3}{51}\times\frac{4}{50}\right)\left(\frac{3!}{2!}\right)$ or $\frac{{}^4C_2\,{}^4C_1}{{}^{52}C_3}$

4. From a pack of 52 cards 3 cards are drawn one by one with replacement. Find the probability that :

(i) all three are diamonds (ii) all three are king

(iii) first is king, second is queen and third is jack (iv) one is king, one is queen and one is jack

(v) first two are kings and third is queen (vi) there are two kings and one queen

Ans. (i) $\left(\frac{13}{52}\right)^3$ (ii) $\left(\frac{4}{52}\right)^3$ (iii) $\left(\frac{4}{52}\right)^3$ (iv) $\left(\frac{4}{52}\right)^3(3!)$ (v) $\left(\frac{4}{52}\right)^3$ (vi) $\left(\frac{4}{52}\right)^3\left(\frac{3!}{2!}\right)$

5. A bag contains 2 red, 3 green and 4 black balls. Three balls are drawn 1 by 1 randomly. What is the probability that
(i) They are of same colour
(a) Without replacement (b) With replacement
(ii) They are of different colours
(a) Without replacement (b) With replacement

Ans. (i) (a) $\frac{{}^3C_3 + {}^4C_3}{{}^9C_3}$ (b) $\left(\frac{2}{9}\right)^3 + \left(\frac{3}{9}\right)^3 + \left(\frac{4}{9}\right)^3$ (ii) (a) $\frac{{}^2C_1\,{}^3C_1\,{}^4C_1}{{}^9C_3}$ (ii) $\left(\frac{2}{9}\times\frac{3}{9}\times\frac{4}{9}\right)(3!)$

6. A purse contains 10 tickets, five printed with I and five printed with T. 3 tickets are drawn without replacement and arranged in the same order in which they are drawn on the table. Find the probability that IIT is formed.

Ans. $\frac{5}{10}\times\frac{4}{9}\times\frac{5}{8}$

7. A bag contains 12 objects. Out of which 4 are defective. Objects are drawn one by one without replacement. Find the probability that first three are non-defective

Ans. $\frac{{}^8C_3}{{}^{12}C_3}$

8. In a bag there are 20 cards. Each card has one of the numbers 1,2,3,4 or 5 painted on it with four cards of each number. 10 cards are drawn from the bag at random. Find the probability that each no. will appear twice.
(i) without replacement (ii) with replacement

Ans. (i) $\frac{{}^4C_2\,{}^4C_2\,{}^4C_2\,{}^4C_2}{{}^{20}C_{10}}$ (ii) $\left(\frac{4}{20}\right)^{10}\left(\frac{10!}{(2!)^5}\right)$

9. Cards are drawn one by one from a pack of 52 cards without replacement. Find the probability that :
(i) no king in first 10 cards (ii) at least one king in first six cards
(iii) at least two kings in first six cards (iv) 7th card is the first king drawn
(v) 15th card is third king drawn

Ans. (i) $\frac{{}^{48}C_{10}}{{}^{52}C_{10}}$ (ii) $1-\frac{{}^{48}C_6}{{}^{52}C_6}$ (iii) $1-\left(\frac{{}^{48}C_6}{{}^{52}C_6}+\frac{{}^{48}C_5\,{}^4C_1}{{}^{52}C_6}\right)$ (iv) $\frac{{}^{48}C_6}{{}^{52}C_6}\times\frac{4}{46}$ (v) $\frac{{}^{48}C_{12}\,{}^4C_2}{{}^{52}C_4}\times\frac{2}{38}$

10. Cards are drawn one by one from a pack of 52 cards with replacement. Find the probability that :
(i) no king in first 10 cards (ii) at least one king in first six cards
(iii) at least two kings in first six cards (iv) 7th card is the first king drawn
(v) 15th card is third king drawn

Ans. (i) $\left(\frac{48}{52}\right)^{10}$ (ii) $1-\left(\frac{48}{52}\right)^6$ (iii) $1-\left(\left(\frac{48}{52}\right)^6+\left(\frac{48}{52}\right)^5\times\frac{4}{52}\times\frac{6!}{5!}\right)$

(iv) $\left(\frac{48}{52}\right)^6\times\left(\frac{4}{52}\right)$ (v) $\left(\left(\frac{48}{52}\right)^{12}\left(\frac{4}{52}\right)^2\right)\frac{14!}{12!2!}\times\frac{4}{52}$

6. INDEPENDENT EVENTS

If two events are such that occurence or non-occurence of one does not affect the chances of occurence or non-occurence of the other event, then the events are said to be independent.

Mathematically :

If A and B are independent, then $P(A / B) = P(A)$.

if $P(A \cap B) = P(A)\, P(B)$, then A and B are independent.

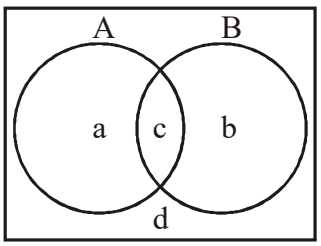
Note: (i) If A and B are independent, then

(a) A′ and B′ are independent,

(b) A and B′ are independent and

(c) A′ and B are independent.

$P(A) = P_1$ $P(B) = P_2$ & A & B are independent



$c = P(AB) = P(A).\, P(B) = P_1P_2$

$b = P(\bar{A}B) = P(\bar{A}).\, P(B) = (1-P_1)P_2$

$a = P(A\bar{B}) = P(A).\ P(\bar{B}) = P_1\,(1-P_2)$

$d = P(\bar{A}.\bar{B}) = P(\bar{A}).P(\bar{B}) = (1-P_2)(1-P_2)$

Three events A, B & C are independent if & only if all the following conditions hold ;

$P(A \cap B) = P(A)\,.\,P(B)$; $P(B \cap C) = P(B).\, P(C)$

$P(C \cap A) = P(C)\,.\,P(A)$ & $P(A \cap B \cap C) = P(A)\,.\,P(B)\,.\,P(C)$

Ex.1 Check whether following events are independent.

(i) A dice is thrown. Event A : Number is odd; Event B : Number is even

(ii) A card is drawn from a pack of 52 cards
Event A : Card is King ; Event B : Card is diamond

(iii) A coin is tossed 3 times
Event A : Getting 3H or 3T; Event B : Getting atleast 2 H; Event C: Getting atmost 2H

(iv) A pair of fair coins. Consider the events:
A = {head on first coin} B = {head on second coin} C = {head on exactly one coin}

Sol. (i) A = {1,3,5}
B = {2,4,6}

$$P(A) = \frac{3}{6} = \frac{1}{2}$$

$P(B) = \frac{3}{6} = \frac{1}{2}$

$P(A \cap B) = 0 \neq P(A) \cdot P(B)$

Not independent

(ii) $P(A) = \frac{4}{52} = \frac{1}{13}$

$P(B) = \frac{13}{52} = \frac{1}{4}$

$P(A \cap B) = \frac{1}{52} = P(A) \cdot P(B)$

Independent

(iii) A = {HHH,TTT}

B = {HHT, HTH, THH, HHH}

C = {HHT, HTH, THH, TTH, THT, HTT, TTT}

$P(A) = \frac{2}{8} = \frac{1}{4}$

$P(B) = \frac{4}{8} = \frac{1}{2}$

$P(C) = \frac{7}{8}$

(1) $P(A \cap B) = \frac{1}{8} = P(A) \cdot P(B) \rightarrow$ A & B independent

$P(B \cap C) = \frac{3}{8} \neq P(B) \cdot P(C) \rightarrow$ Not independent

$P(C \cap A) = \frac{7}{8} \neq P(C) \cdot P(A) \rightarrow$ Not independent

(iv) A = {HH, HT}

B = {HH, TH}

C= {HT, TH}

$P(A) = \frac{2}{4} = \frac{1}{2}$

$P(B) = \frac{1}{2}$

$P(C) = \frac{1}{2}$

(1) $P(A \cap B) = \frac{1}{4} = P(A)\ P(B) \rightarrow$ Yes

$P(B \cap C) = \frac{1}{4} = P(B)\ P(C) \rightarrow$ Yes

$P(C \cap A) = \frac{1}{4} = P(C) \cdot P(A) \rightarrow$ Yes

$P(A \cap B \cap C) = 0 \neq P(A) \cdot P(B) \cdot P(C) \rightarrow$ No

Questions on Independent Events

1. A problem in mathematics is given to 2 children who solve it independently. If probability of A solving it is 1/2 and probability of B solving it is 2/3. Find the probability that the problem is solved.

Ans. 5/6

Sol. $P(A) = \frac{1}{2}$

$P(B) = \frac{2}{3}$

$P(A\overline{B}) + P(\overline{A}B) + P(AB)$

$= \frac{1}{2} \times \frac{1}{3} + \frac{1}{2} \times \frac{2}{3} + \frac{1}{2} \times \frac{2}{3} = \frac{5}{6}$

M-II : $P = 1 - P(\overline{A}\overline{B})$

$= 1 - \frac{1}{2} \times \frac{1}{3} = \frac{5}{6}$

2. Probability that a teacher takes a surprise test is 1/3. If a student remains absent for two days then find the probability that he misses :

(i) exactly one test (ii) at least one test (iii) atmost one test

Ans. (i) 4/9 (ii) 5/9 (iii) 8/9

Sol. $P(T) = \frac{1}{3}$

$P(\overline{T}) = \frac{2}{3}$

(i) P(Exactly 1 Test) = $P(T\overline{T}) + P(\overline{T}T)$

$= \frac{1}{3} \times \frac{2}{3} + \frac{2}{3} \times \frac{1}{3} = \frac{4}{9}$

(ii) P(atleast 1 Test) = 1 – P(0 test miss)

$= 1 - P(\overline{T}\overline{T})$

$= 1 - \frac{2}{3} \times \frac{2}{3} = \frac{5}{9}$

(iii) P(atmost 1 test) = P(0 test miss) + P(1 test miss)

$$= P\left(\overline{T}\,\overline{T}\right) + P\left(T\overline{T}\right) + P\left(\overline{T}T\right)$$

$$= \frac{2}{3}\times\frac{2}{3} + \frac{1}{3}\times\frac{2}{3} + \frac{2}{3}\times\frac{1}{3} = \frac{8}{9}$$

3. A coin whose faces are marked with 1 and 2 is tossed 4 times. Find the probability that :

(i) total is 6 (ii) total is at least five (iii) total is 5 (iv) total is 5 given that total is at least 5

Ans. (i) 6/16 (ii) 15/16 (iii) 4/16 (iv) 4/15

Sol. (i) 1 1 2 2

$$= \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{4!}{2!2!}$$

$$= \frac{6}{16}$$

(ii) P(5) + P(6) + P(7) + P(8)
1 1 1 2 1 1 2 2 1 2 2 2 2 2 2 2

$$\left(\frac{1}{2}\right)^4\times\frac{4!}{3!} + \left(\frac{1}{2}\right)^4\times\frac{4!}{2!2!} + \left(\frac{1}{2}\right)^4\times\frac{4!}{3!} + \left(\frac{1}{2}\right)^4$$

$$= \frac{15}{16}$$

M-II $\Rightarrow$ 1 – P(4)
1 1 1 1

$$\Rightarrow 1 - \left(\frac{1}{2}\right)^4 = \frac{15}{16}$$

(iii) P(5)
1112

$$\Rightarrow \left(\frac{1}{2}\right)^4\times\frac{4!}{3!} = \frac{4}{16}$$

(iv) $$P(E) = \frac{P(5)}{P(5)+P(6)+P(7)+P(8)} = \frac{\frac{4}{16}}{\frac{15}{16}} = \frac{4}{15}$$

4. A dice is thrown 5 times what is the probability that number 3 appears exactly twice

Ans. $$\frac{5!}{2!3!}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^3$$

5. Suppose a student A attends class 80% of the time and B attends class 60% of the time. The absences of the two students are independent.

(i) Find the probability that at least one of the students will be present in the class on a given day.

(ii) If exactly one of the students A and B is present in the class, find the probability that A is present.

(iii) If at least one of the students A or B is present in the class on a given day, find the probability that A is present.

Ans. (i) $\frac{23}{25}$ (ii) $\frac{8}{11}$ (iii) $\frac{20}{23}$

Sol. (i) $P\left(A\overline{B}\right)+P\left(\overline{A}B\right)+P\left(AB\right)$

$$\frac{4}{5}\times\frac{2}{5}+\frac{1}{5}\times\frac{3}{5}+\frac{4}{5}\times\frac{3}{5}=\frac{23}{25}$$

OR

$$1-\frac{1}{5}\times\frac{2}{5}=\frac{23}{25}$$

(ii) $$\frac{P\left(A\overline{B}\right)}{P\left(A\overline{B}\right)+P\left(\overline{A}B\right)}=\frac{\frac{4}{5}\times\frac{2}{5}}{\frac{4}{5}\times\frac{2}{5}+\frac{1}{5}\times\frac{3}{5}}=\frac{8}{11}$$

(iii) $$\frac{P\left(A\overline{B}\right)+P\left(AB\right)}{P\left(A\overline{B}\right)+P\left(\overline{A}B\right)+P\left(AB\right)}=\frac{20}{23}$$

6. $P(A)=\frac{1}{3}$, $P(B)=\frac{1}{2}$, find $P(A\cap B)$ if :

(i) A and B are independent (ii) A and B are disjoint (iii) $A\Rightarrow B$

(iv) $P(A/B)=\frac{1}{8}$ (v) $P(AB)=\frac{1}{8}$

Ans. (i) 1/6 (ii) 0 (iii) 1/3 (iv) 1/16 (v) 1/8

Sol. (i) $P(A\cap B)=P(A)\cdot P(B)$

$$=\frac{1}{3}\times\frac{1}{2}$$

$$=\frac{1}{6}$$

(ii) $P(A\cap B)=0$

(iii) $A\Rightarrow B$

$$P\left(A\cap B\right)=P\left(A\right)=\frac{1}{3}$$

(iv) $$P\left(\frac{A}{B}\right)=\frac{1}{8}$$

$$\frac{P\left(A\cap B\right)}{P\left(B\right)}=\frac{1}{8}$$

$P(A \cap B) = \frac{1}{16}$

(v) $P(AB) = \frac{1}{8}$

$P(A \cap B) = \frac{1}{8}$

7. A box contains 5 tubes, 2 of them defective and 3 good one. Tubes are tested by one-by-one till the 2 efective tubes are discovered. What is the probability that the testing procedure comes to an end at the end of

(A) second testing (B) 3rd testing

Ans. (A) 1/10 (B) 3/10

Sol. (A) 1st 2nd

defective & defective

$\frac{2}{5} \times \frac{1}{4} = \frac{1}{10}$

(B) Case 1 : good, defective, defective $= \frac{3}{5} \times \frac{2}{4} \times \frac{1}{3}$

Case 2 : defective, good, defective $= \frac{2}{5} \times \frac{3}{4} \times \frac{1}{3}$

Case 3 : good, good, good $= \frac{3}{5} \times \frac{2}{4} \times \frac{1}{3}$

Total $= \frac{3}{10}$

8. All face cards from a pack of 52 playing cards are removed. From the remaining 40 cards, 4 are drawn. Find the probability

(a) All the four cards are of the same denominations.

(b) that they are two cards of one denominations and two cards are of other denominations.

(c) that they are of different suit & different denominations.

Ans. (a) $\frac{{}^{10}C_1 {}^{4}C_4}{{}^{10}C_4}$ (b) $\frac{{}^{10}C_2 {}^{4}C_2 {}^{4}C_2}{{}^{40}C_4}$ (c) $\frac{{}^{10}C_4 {}^{4}C_1 {}^{3}C_1 {}^{2}C_1 {}^{1}C_1}{{}^{40}C_4}$

9. Ram and Shyam toss a dice according to their turn indefinitely while the face 1 appears. The person getting the face 1 for the first time wins the game. If ram starts the game then find their respective change of winning.

Ans. P (Ram) = 6/11 P (Shyam) = 5/11

Sol. P(Ram wins) $= P(1) + P(\bar{1}\bar{1}1) + P(\bar{1}\bar{1}\bar{1}\bar{1}1) + ...$

$= \frac{1}{6} + \left(\frac{5}{6}\right)^2 \times \frac{1}{6} + \left(\frac{5}{6}\right)^4 \times \frac{1}{6} + ...$

$$= \frac{\frac{1}{6}}{1-\left(\frac{5}{6}\right)^2} = \frac{6}{11}$$

P(Shyam wins) = 1 – P(Ram wins)

$$= 1 - \frac{6}{11} = \frac{5}{11}$$

M-II : P(Ram) + P(Shyam) = 1

$$x + \frac{5}{6}x = 1$$

$$x = \frac{6}{11}$$

10. Ram, Shayam and Ghanshyam toss a dice according to their turn to infinitely while the face 1 appears. Person getting the face 1 first time wins the game. If ram starts the game then find their respective changes of winning.

Ans. P(Ram) = 36/91 ; P (Shyam) = 30/91 ; P(Ghanshyam) = 25/91

Sol. $P(\text{Ram wins}) = \left\{P(1) + P(\bar{1}\bar{1}\bar{1}1) + P(\bar{1}\bar{1}\bar{1}\bar{1}\bar{1}\bar{1}1) + ...\right\}$

$$= \frac{1}{6} + \left(\frac{5}{6}\right)^3 \times \frac{1}{6} + \left(\frac{5}{6}\right)^6 \times \frac{1}{6} + ...$$

$$= \frac{\frac{1}{6}}{1-\left(\frac{5}{6}\right)^3} = \frac{36}{91}$$

$P(\text{Shyam wins}) = P(\bar{1}1) + P(\bar{1}\bar{1}\bar{1}\bar{1}1) + P(\bar{1}\bar{1}\bar{1}\bar{1}\bar{1}\bar{1}\bar{1}1)$

$$= \frac{5}{6} \times \frac{1}{6} + \left(\frac{5}{6}\right)^4 \times \frac{1}{6} + \left(\frac{5}{6}\right)^7 \times \frac{1}{6} + ...$$

$$= \frac{30}{91}$$

P(Ghanshyam wins) = 1 – P(Ram) – P(Shyam)

$$= 1 - \frac{36}{91} - \frac{30}{91} = \frac{25}{91}$$

M-II : P(Ram) + P(Shyam) + P(Ghanshyam) = 1

$$x + \frac{5}{6}x + \frac{5}{6} \times \frac{5}{6}x = 1$$

$$x = \frac{36}{91}$$

11. Ram, Shyam and Ghanshyam toss a coin according to their indefinitely untill H appears. Person getting head for first time wins the game. If ram starts the game find their respective chances of winning.

Ans. P(R) = 4/7; P(S) = 2/7 ; P (G) = 1/7

Sol. P(Ram) + P(Shyam) + P(Ghanshyam) = 1

$$x + \frac{1}{2}x + \frac{1}{2}\times\frac{1}{2}x = 1$$

$$x = \frac{4}{7}$$

12. An urn contains 1R , 2G and 3B balls. Three people A, B & C in order draw one ball from the urn and put it back after noting its colour. They continue doing it indefinitely unless one who draws a red ball first wins the game. Compute their respective chances of winning the game.

Ans. P(A) = 36/91; P(B) = 30/91 ; P (C) = 25/91

13. Two persons A and B one by one in order drawn one ball each from a purse containing 5W and 1R balls and retain it. The person who gets a red ball wins the game.
E : event that 'A' wins and F : event that 'B' wins, then E and F are equiprobable. [T/F]

Ans. $P(A) = \frac{1}{2} = P(B)$

Sol. P(E) = P(R) + P(WWR) + P(WWWWR)

$$= \frac{1}{6} + \frac{5}{6}\times\frac{4}{5}\times\frac{1}{4} + \frac{5}{6}\times\frac{4}{5}\times\frac{3}{4}\times\frac{2}{3}\times\frac{1}{2}$$

$$= \frac{1}{2}$$

P(F) = 1 – P(E)

$$= \frac{1}{2}$$

14. In the following experiment, we roll a fair die 5 times
(a) What is the probability of the sequence "1, 2, 3, 4, 5".
(b) What is the probability that the sequence starts with a "1"
(c) What is the probability that the number "2" appears exactly twice.
(d) Let E be the event that we find the sequence "1, 2, 3, 4, 5" and let F be the event that the sequence starts with a "1". What are the probabilities P(E/F) and P(F/E)

Ans. (a) $\left(\frac{1}{6}\right)^5$ (b) $\frac{1}{6}$ (c) $\frac{5!}{2!3!}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^3$ (d) $P(E/F) = \left(\frac{1}{6}\right)^4$; $P(F/E) = 1$

7. LAW OF TOTAL PROBABILITY

If an event A can occur with one of the n mutually exclusive and exhaustive events B_1, B_2, ..., B_n and

the probabilities $P(A/B_1)$, $P(A/B_2)$ .... $P(A/B_n)$ are known, then

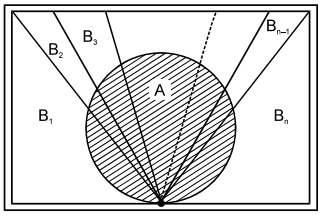
$$P(A) = \sum_{i=1}^{n} P(B_i).P(A/B_i)$$

Proof :

The event A occurs with one of the n mutually exclusive and exhaustive events

$B_1, B_2, B_3, \ldots\ldots, B_n$

$A = (A \cap B_1) \cup (A \cap B_2) \cup (A \cap B_3) \cup \ldots\ldots\ldots \cup (A \cap B_n)$



$$P(A) = P(A \cap B_1) + P(A \cap B_2) + \ldots\ldots + P(A \cap B_n) = \sum_{i=1}^{n} P(A \cap B_i)$$

Now,

$P(A \cap B_i) = P(A) \cdot P(B_i/A) = P(B_i) \cdot P(A/B_i)$

$$\therefore \quad P(A) = \sum_{i=1}^{n} P(B_i).P(A/B_i)$$

Questions on Law of Total Probability

1. A girl has 3 compartments in her purse. First compartment contains 1R and 2P coins. Second compartment contatins 2R and 3P coins. Third compartment contains 3R and 4P coins. She randomly selects a compartment and takes out a coin. What is the probability that it is a Rupee coin ?

Ans. $\frac{122}{315}$

Sol.

1st	2nd	3rd
1R	2R	3R
2P	3P	4P

$P(R) = P(1^{st} \& R) + P(2^{nd} \& R) + P(3^{rd} \& R)$

$$\frac{1}{3}\times\frac{1}{3}+\frac{1}{3}\times\frac{2}{5}+\frac{1}{3}\times\frac{3}{7}$$

$$=\frac{122}{315}$$

2. In a bag there are 3 unbiased coin and one double headed (DH) coin. A coin is randomly drawn from the bag and tossed. Find the probability that head appears.

Ans. $\frac{5}{8}$

Sol. $P(H) = P(\text{unbiased coin} \& H) + P(DH \& H)$

$$=\frac{3}{4}\times\frac{1}{2}+\frac{1}{4}\times 1$$

$$=\frac{5}{8}$$

3. In a bag there are 3 unbiased coins, 2 DH coins and one DT coin. A coin is randomly drawn fromthe bag and tossed. Find the probability that head appears.

Ans. $\frac{7}{12}$

Sol. P(H) = P(Unbiased & H) + P(DH & H) + P(DT & H)

$$=\frac{3}{6}\times\frac{1}{2}+\frac{2}{6}\times 1+0$$

$$=\frac{7}{12}$$

4. Three political parties namely the B.J.P., the Congress and the Janta Dal are contesting for a state legislative assembly elections. The state does not have a common entrance test after the 12th standard, for the admissions to the medical or engineering colleges. The probabilities of these parties winning the elections are $\frac{1}{3},\frac{4}{9}$ and $\frac{2}{9}$ respectively. If the B.J.P. comes to the power, the probability of its introducing the common entrance test for the state is 0.6 and the corresponding probabilities for theCongress and Janta Dal are 0.7 and 0.5 respectively. Find the probability that the common entrance test is introduced.

Ans. $\frac{28}{45}$

Sol. P(Entrance Exam) = P(Congress & Exam) + P(BJP & Exam) + P(JD & Exam)

$$=\frac{4}{9}\times 0.7+\frac{1}{3}\times 0.6+\frac{2}{9}\times 0.5$$

$$=\frac{28}{45}$$

5. In a company there are 3 machines x, y and z. Machine x produces 50% of the total production out of which 3% is defective. Machine y produces 30% of the total production out of which 4% is defective and z produces remaining production out of which 5% is defective. An object is randomly selected from the total production. Find the probability that it is defective.

Ans. $\frac{37}{1000}$

Sol. P(Defective) = P(x & defective) + P(y & defective) + P(z & defective)

$$=\frac{50}{100}\times\frac{3}{100}+\frac{30}{100}\times\frac{4}{100}+\frac{20}{100}\times\frac{5}{100}$$

$$=\frac{37}{1000}$$

6. A lot contains 20 articles. The probability that the lot contains exactly 2 defective articles is 0.4 & the probability that the lot contains exactly 3 defective articles is 0.6. Articles are drawn from the lot at random one by one without replacement and are tested till all defective articles are found. The probability that the testing procedure ends at the twelfth testing **[JEE 86]**

Ans. $(0.4)\left(\frac{{}^{18}C_{10}\,{}^{2}C_{1}}{{}^{20}C_{11}}\times\frac{1}{9}\right)+(0.6)\left(\frac{{}^{17}C_{9}\,{}^{13}C_{2}}{{}^{20}C_{11}}\times\frac{1}{9}\right)$

7. In a bag there are 2 unbiased coins and 2DH coins and 1 DT coin, Coin is randomly drawn from the bag and tossed.

(i) Find the probability that head appears

(ii) It is given that head appears then find the probability that it was a DH coin

Ans. (i) 3/5 (ii) 2/3

Sol. (i) P(H) = P(Unbiased & H) + P(DH & H) + P(DT & H)

$=\frac{2}{5}\times\frac{1}{2}+\frac{2}{5}\times1+0$

$=\frac{3}{5}$

(ii) $\frac{P(DH\ \&\ H)}{P(\text{Unbiased}\ \&\ H)+P(DH\ \&\ H)+P(DT\ \&\ H)}$

$=\frac{\frac{2}{5}\times1}{\frac{2}{5}\times\frac{1}{2}+\frac{2}{5}\times1+0}$

8. BAYE'S THEOREM

If an event A can occur with one of the n mutually exclusive and exhaustive events $B_1, B_2, \ldots, B_n$ and the probabilities $P(A/B_1), P(A/B_2) \ldots P(A/B_n)$ are known, then

$$P(B_i/A)=\frac{P(B_i).P(A/B_i)}{\sum_{i=1}^{n}P(B_i).P(A/B_i)}$$

Proof:

The event A occurs with one of the n mutually exclusive and exhaustive events

$B_1, B_2, B_3, \ldots\ldots, B_n$

$A = (A \cap B_1) \cup (A \cap B_2) \cup (A \cap B_3) \cup \ldots\ldots\ldots \cup (A \cap B_n)$

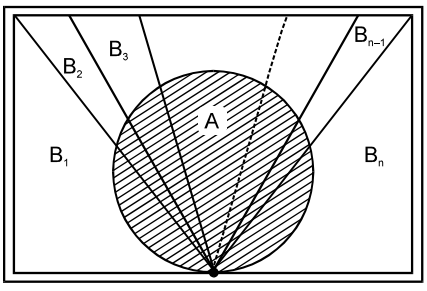
$P(A) = P(A \cap B_1) + P(A \cap B_2) + \ldots\ldots\ldots + P(A \cap B_n) = \sum_{i=1}^{n}P(A \cap B_i)$

Now,

$P(A \cap B_i) = P(A)\ .\ P(B_i/A) = P(B_i)\ .\ P(A/B_i)$

$P\ (B_i/A) = \frac{P(B_i).P(A/B_i)}{P(A)} = \frac{P(B_i).P(A/B_i)}{\sum_{i=1}^{n}P(A \cap B_i)}$

$P(B_i/A) = \frac{P(B_i).P(A/B_i)}{\sum P(B_i).P(A/B_i)}$



Questions on Baye's Theorem

1. In a bag there are 4 unbiased coin and 1DH coin one coin is randomly drawn and tossed twice. If head appears both the times then find the probability that it was an unbaised coin.

Ans. 1/2

Sol. $$P=\frac{P(\text{Unbiased}\rightarrow H\,\&\,H)}{P(\text{Unbiased}\rightarrow H\,\&\,H)+P(\text{DH}\rightarrow H\,\&\,H)}$$

$$P=\frac{\frac{4}{5}\times\frac{1}{2}\times\frac{1}{2}}{\frac{4}{5}\times\frac{1}{2}\times\frac{1}{2}+\frac{1}{5}\times1\times1}=\frac{1}{2}$$

2. A girl has 3 compartments in her purse. First compartment contains 1R and 2P coins. Second compartment contatins 2R and 3P coins. Third compartment contains 3R and 4P coins. She randomly selects a compartment and takes out a coin. If it is a Re coin then find the probability that it is from comp. C_3.

Ans. 45/122

Sol. $$P=\frac{P(3^{rd}\rightarrow R)}{P(1^{st}\rightarrow R)+P(2^{nd}\rightarrow R)+P(3^{rd}\rightarrow R)}$$

$$P=\frac{\frac{1}{3}\times\frac{3}{7}}{\frac{1}{3}\times\frac{1}{3}+\frac{1}{3}\times\frac{2}{5}+\frac{1}{3}\times\frac{3}{7}}=\frac{45}{122}$$

3. A bag contains 3 cards one of them is red on both sides, other is blue on both sides and third one is red on one and blue on other side card is randomly drawn and put on the table. Its upper face shown red colour. Find the probability that other face will also be red.

Ans. 2/3

Sol. $\boxed{1^{st}\ R/R}\quad\boxed{2^{nd}\ B/B}\quad\boxed{3^{rd}\ R/B}$

$$P=\frac{P(1^{st}\rightarrow R)}{P(1^{st}\rightarrow R)+P(2^{nd}\rightarrow R)+P(3^{rd}\rightarrow R)}$$

$$P=\frac{\frac{1}{3}\times1}{\frac{1}{3}\times1+\frac{1}{3}\times0+\frac{1}{3}\times\frac{1}{2}}=\frac{2}{3}$$

4. Suppose we have 10 coins such that if the ith coin is flipped, heads will appear with probability $\frac{i}{10}$. When one of the coin is randomly selected and flipped, it shows up head. The probability that it was the fifth coin is.

Ans. 1/11

Sol. $$P=\frac{P(C_5 \to H)}{P(C_1 \to H)+P(C_2 \to H)+.....+P(C_{10} \to H)}$$

$$P=\frac{\frac{1}{10}\times\frac{5}{10}}{\frac{1}{10}\times\frac{1}{10}+\frac{1}{10}\times\frac{2}{10}+....+\frac{1}{10}\times\frac{10}{10}}$$

$$P=\frac{1}{11}$$

5. 'A' writes a letter to his friend and gives it to his son to post it in a letter box, the reliability of his son being 3/4. The probability that a letter posted will get delivered is 8/9. At a later date 'A' hears from 'B' that the letter has not reached him. Find the probability that the son did not post the letter at all.

Ans. 3/4

Sol. $$P=\frac{P(\text{son did not post})}{P(\text{son posted but post office did not deliver})+P(\text{son did not post})}$$

$$P=\frac{\frac{1}{4}\times 1}{\frac{3}{4}\times\frac{1}{9}+\frac{1}{4}\times 1}$$

$$P=\frac{3}{4}$$

6. A bag contains 5 balls of unknown colour. All possible no. of red balls in the bag are equally likly. Balls are randomly drawn twice with replacement and they were found to be red on both occassions. Then find the probability that there were 3 red balls in the bag.

Ans. 9/55

Sol. P(0 Red) = P(1 Red) = P(2 Red) = P(3 Red) = P(4 Red) = P(5 Red) = $\frac{1}{6}$

$$P=\frac{P(3R \to R \,\&\, R)}{P(0R \to R \,\&\, R)+P(1R \to R \,\&\, R)+.....+P(5R \to R \,\&\, R)}$$

$$P=\frac{\frac{1}{6}\times\frac{3}{5}\times\frac{3}{5}}{\frac{1}{6}\times 0+\frac{1}{6}\times\frac{1}{5}\times\frac{1}{5}+\frac{1}{6}\times\frac{2}{5}\times\frac{2}{5}+.....+\frac{1}{6}\times\frac{5}{5}\times\frac{5}{5}}$$

$$= \frac{9}{55}$$

7. In a bag there are 3 unbaised coins and 2DH coins. Coin is randomly drawn and tossed and it appears heads the same coin is tossed again then find the probability that it will appear heads.

Ans. 11/14

Sol. $$P = \frac{P(\text{Unbiased} \to H\&H) + P(DH \to H \& H)}{P(\text{Unbiased} \to H) + P(DH \to H)}$$

$$\frac{\frac{3}{5}\times\frac{1}{2}\times\frac{1}{2} + \frac{2}{5}\times 1\times 1}{\frac{3}{5}\times\frac{1}{2} + \frac{2}{5}\times 1} = \frac{11}{14}$$

8. A bag contains 5 balls of unknown colours. All possible no. of red balls in the bag are equally likely. Balls are randomly drawn twice with replacement and they were found to be red on both ocassions. Now a ball is again drawn then find the probability that it is a red ball.

Ans. 9/11

Sol. P(0 Red) = P(1 Red) = P(2 Red) = P(3 Red) = P(4 Red) = P(5 Red) = $\frac{1}{6}$

$$P = \frac{P(0R \Rightarrow R \& R \to R) + P(1R \to R \& R \to R) + ...P(5R \to R \& R \to R)}{P(0\,Red \to R \& R) + P(1\,Red \to R \& R) + ...P(5R \to R \& R)}$$

$$= \frac{\frac{1}{6}\times 0 + \frac{1}{6}\times\frac{1}{5}\times\frac{1}{5}\times\frac{1}{5} + ... + \frac{1}{6}\times\frac{5}{5}\times\frac{5}{5}\times\frac{5}{5}}{\frac{1}{6}\times 0 + \frac{1}{6}\times\frac{1}{5}\times\frac{1}{5} + ... + \frac{1}{6}\times\frac{5}{5}\times\frac{5}{5}}$$

$$= \frac{9}{11}$$

9. Atrial consists of rolling a red die and a blue die the dice being fair. The result R of the trial is defined as the sum of the two numbers showing when the numbers on the red and the blue dice are the same but as the product of these two numbers when they are different.

(i) The probability that the result of a throw is 12, is

(A) $\frac{1}{12}$ (B) $\frac{1}{9}$ (C) $\frac{5}{36}$ (D) $\frac{1}{6}$

(ii) If $R \geq 15$, then $P(R \geq 20)$ is

(A) 3/5 (B) 2/5 (C) 1/2 (D) 1/3

(iii) If the result of two such throws are added then $P(R \geq 45)$

(A) $\frac{5}{108}$ (B) $\frac{5}{648}$ (C) $\frac{5}{162}$ (D) None of these

Ans. (i) C (ii) A (iii) $\frac{5}{162}$

9. BINOMIAL PROBABILITY DISTRIBUTION

Bernoulli Trials :- Trials of random experiment are called bernoulli trials if they satisfy following condi tions

(i) These should be finite number of trials.

(ii) Trials should be independent of each other.

(iii) Each trial should have two outcomes- success or failure

(iv) Probability of success remains same in each trial.

Let p be the probability that an event will happen in any single Bernoulli trial (called the probability of success). Then q = 1 – p is the probability that the event will fail to happen in any single trial (called the probability of failure). The probability that the event will happen exactly r times in n trials (i.e., r successes and n – r failures will occur) is given by the probability function.

$$P(X = r) = {}^{n}C_{r}\, p^{r}(1-p)^{n-r} = \frac{n!}{r!(n-r)!}\, p^{r}(1-p)^{n-r} \quad \text{..... (i)}$$

where the random variable X denotes the number of successes in n trials and r = 0, 1,....., n.

This distribution is denoted as B(n,p)

Questions on Binomial Probability Distribution

1. Coin is tossed 100 times. Find probability of getting exactly 30 heads.

Ans. $^{100}C_{30}\left(\frac{1}{2}\right)^{100}$

2. A dice is thrown 50 times. Find the probability of getting 6 total 10 times :

Ans. $^{50}C_{10}\left(\frac{1}{6}\right)^{10}\left(\frac{5}{6}\right)^{40}$

3. A pair of dice is thrown 6 times, getting a doublet is considered a success. Compute the probability of

(a) no success (b) exactly one success

(c) at least one success (d) at most one success

Ans. (a) $^{6}C_{0}\left(\frac{5}{6}\right)^{6}$ (b) $^{6}C_{1}\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^{5}$ (c) $1 - {}^{6}C_{0}\left(\frac{5}{6}\right)^{6}$ (d) $^{6}C_{0}\left(\frac{5}{6}\right)^{6} + {}^{6}C_{1}\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^{6}$

4. A coin is twice as likely to land heads as tails. In a sequence of five independent trials, find the probability that the third head occurs on the fifth toss.

Ans. $\frac{16}{81}$

Sol. P(H) = 2P(T)
P(H) + P(T) = 1
2P(T) + P(T) = 1

$$P(T)=\frac{1}{3}\ \&\ P(H)=\frac{2}{3}$$

$$\Rightarrow\ {}^4C_2\left(\frac{2}{3}\right)^2\left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right)=\frac{16}{81}$$

5. In a hurdle race a man has to clear 9 hurdles. Probability that he clears a hurdle 2/3 and the probability that he knocks down the hurdle is 1/3. Find the probability that he knocks down fewer than 2 hurdles.

Ans. $${}^9C_8\left(\frac{2}{3}\right)^8\left(\frac{1}{3}\right)+{}^9C_9\left(\frac{2}{3}\right)^9$$

6. A drunkard takes a step forward or backward. The probability that he takes a step forward is 0.4. Find the probability that at the end of 11 steps he is one step away from the starting point. **[JEE 87]**

Ans. $${}^{11}C_5\left(\frac{2}{5}\right)^5\left(\frac{3}{5}\right)^6+{}^{11}C_6\left(\frac{2}{5}\right)^6\left(\frac{3}{5}\right)^5$$

7. 100 identical coins each falling head wise with probability p (0<p<1) are tossed once . If the probability of 50 coins showing up the heads is equal to the probability of 51 coins showing up the heads, find the value of p. **[JEE 88]**

Ans. $P=\frac{51}{101}$

Sol. $n = 100$

$p(H) = p$

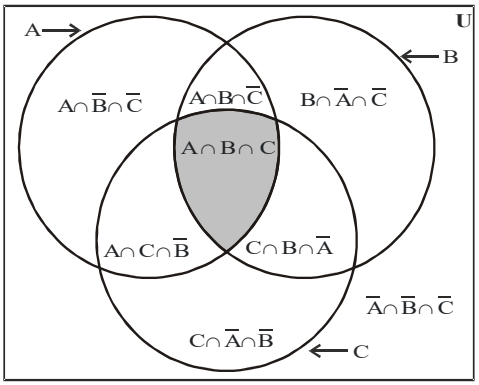
$p(T) = 1 - p$

$p(X = 50) = p(X = 51)$

$${}^{100}C_{50}\,p^{50}\,(1-p)^{50}={}^{100}C_{51}\,p^{51}\,(1-p)^{49}$$

$$P=\frac{51}{101}$$

10.1 VENN DIAGRAM OF 3 EVENTS



For any three events A,B and C we have

(a) $P(A \text{ or } B \text{ or } C) = P(A) + P(B) + P(C) - P(A \cap B)$

$- P(B \cap C) - \quad P(C \cap A) + P(A \cap B \cap C)$

(b) P(at least two of A,B,C occur)

$= P(B \cap C) + P(C \cap A) + P(A \cap B) - 2P(A \cap B \cap C)$

(c) P(exactly two of A,B,C occur)

$= P(B \cap C) + P(C \cap A) + P(A \cap B) - 3P(A \cap B \cap C)$

(d) P(exactly one of A,B,C occurs)

$= P(A) + P(B) + P(C) - 2P(B \cap C) - 2P(C \cap A) - 2P(A \cap B) + 3P(A \cap B \cap C)$

Note : If three events A, B and C are pair wise mutually exclusive then they must be mutually exclusive. i.e $P(A \cap B) = P(B \cap C) = P(C \cap A) = 0 \Rightarrow P(A \cap B \cap C) = 0$. However the converse of this is not true.

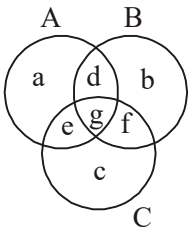
Questions on Venn Diagram of 3 events

1. There are 3 clubs A, B and C in a town with 40, 50 and 60 members respectively. 10 people are members of all the three clubs , 70 are members in only one club. A member is randomly selected. Find the probability that he had membership of exactly two clubs.

(A) $\frac{7}{15}$ (B) $\frac{1}{6}$ (C) $\frac{3}{21}$ (D) $\frac{5}{21}$

Ans. D

Sol.



(i) $g = 10$

(ii) $a + b + c = 70$

$a + d + g + e = 40$

$a + d + e = 30$ ...(3)

$b + d + g + f = 50$

$b + d + f = 40$ ...(4)

$c + e + g + f = 60$

$c + e + f = 50$ ...(5)

(3) + (4) + (5)

$a + b + c + 2(d + e + f) = 120$

$d + e + f = 25$

$$P = \frac{d+e+f}{a+b+c+d+e+f+g}$$

$$P = \frac{25}{70+25+10} = \frac{5}{21}$$

2. Three persons A, B, C independently fire at a target. Suppose $P(A) = \frac{1}{6}$; $P(B) = \frac{1}{4}$; $P(C) = \frac{1}{3}$ denote their probabilities of hitting the target.

(a) Find the probability that at least one of them hits the target.

(b) Find the probability that exactly one of them hits the target.

(c) If the target is hit only once, find the probability that it was the man A.

Ans. (a) 7/12 (b) 31/72 (c) 6/31

Sol. (a) $1 - P\left(\overline{A}\,\overline{B}\,\overline{C}\right)$

$$1-\left(\frac{5}{6}\times\frac{3}{4}\times\frac{2}{3}\right)=\frac{7}{12}$$

(b) $P\left(A\overline{B}\overline{C}\right)+P\left(\overline{A}B\overline{C}\right)+P\left(\overline{A}\overline{B}C\right)$

$$\Rightarrow \frac{1}{6}\times\frac{3}{4}\times\frac{2}{3}+\frac{5}{6}\times\frac{1}{4}\times\frac{2}{3}+\frac{5}{6}\times\frac{3}{4}\times\frac{1}{3}$$

$$=\frac{31}{72}$$

(c) $P(A) + P(B) - 2P(A\cap B) = P$

$P(B) + P(C) - 2P(B\cap C) = P$

$P(C) + P(A) - 2P(C\cap A) = P$

3. For the 3 events A,B and C

P(exactly one of the events A or B occurs)
P(exactly one of the events B or C occurs) } = p
P(exactly one of the events C or A occurs)

P(all the 3 events occur simultaneously) = p^2

If A, B, C are exhaustive then find the value(s) of p.

Ans. $P = \frac{1}{2}$

Sol. $2(P(A) + P(B) + P(C) - P(A\cap B) - P(B\cap C) - P(C\cap A)) = 3P$

$P(A\cap B\cap C) = P^2$

$\Rightarrow A\cup B\cup C = S$

$P(A\cup B\cup C) = 1$

$\Sigma P(A) - \Sigma P(A\cap B)+P(A\cap B\cap C) = 1$

$$\frac{3P}{2}+P^2=1$$

$2P^2 + 3P - 2 = 0$

$$P=\frac{1}{2} \text{ or } -2$$

So, $P=\frac{1}{2}$

10.2 PROBABILITY THROUGH TREE DIAGRAM

These tree diagrams are generally drawn by economist and give a simple approach to solve a problem.

Q.1 A bag initially contains 1 red ball and 2 blue balls. A trial consists of selecting a ball at random, noting its colour and replacing it together with an additional ball of the same colour. Given that three trials are made,draw a tree diagram illustrating the various probabilities. Hence, or otherwise, find the probability that.

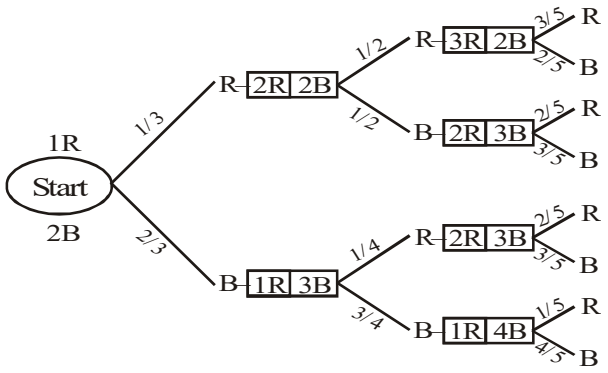
(a) atleast one blue ball is drawn

(b) exactly one blue ball is drawn

(c) Given that all three balls drawn are of the same colour find the probability that they are all red.

Ans.

Sol :



Calculations :

$$P(A) = 1 - P(RRR) = 1 - \frac{1}{3}\cdot\frac{1}{2}\cdot\frac{3}{5} = 1 - \frac{1}{10} = \frac{9}{10}$$

$$P(\text{exactly one Blue}) = \frac{1}{3}\cdot\frac{1}{2}\cdot\frac{2}{5} + \frac{1}{3}\cdot\frac{1}{2}\cdot\frac{2}{5} + \frac{2}{3}\cdot\frac{1}{4}\cdot\frac{2}{5} = \frac{1}{15} + \frac{1}{15} + \frac{1}{15} = \frac{3}{15} = \frac{1}{5}$$

$$P(C) = P\left(\frac{RRR}{(RRR \cup BBB)}\right) = \frac{P(RRR)}{P(RRR) + P(BBB)} = \frac{\frac{1}{3}\cdot\frac{1}{2}\cdot\frac{3}{5}}{\frac{1}{3}\cdot\frac{1}{2}\cdot\frac{3}{5} + \frac{2}{3}\cdot\frac{3}{4}\cdot\frac{4}{5}} = \frac{\frac{1}{10}}{\frac{1}{10} + \frac{4}{10}} = \frac{1}{5}$$

Q.2 Box A contains nine cards numbered 1 through 9, and box B contains five cards numbered 1 through 5. A box is chosen at random and a card drawn; if the card shows an even number, another card is drawn from the same box. If the card shows an odd number, a card is drawn from the other box ;

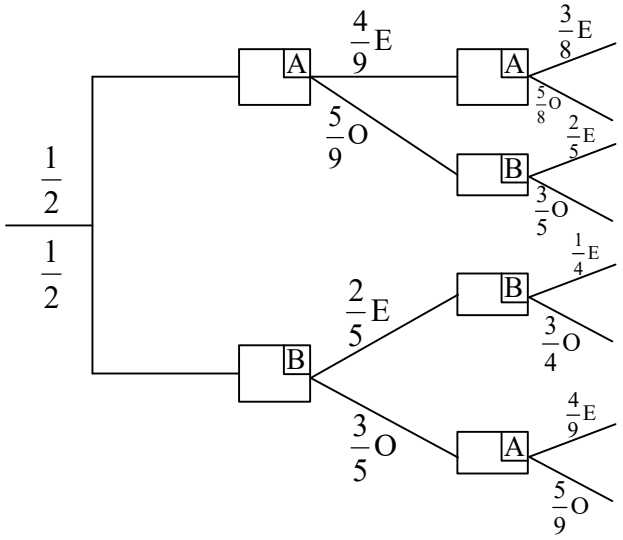
(i) What is the probability that both cards show even numbers?

(ii) If both cards show even numbers, what is the probability that they come from box A?

(iii) What is the probability that both cards show odd numbers?

Ans. 2/15 (ii) 5/8 (iii) 1/3

Sol.



(i) $P(E)=\frac{1}{2}\times\frac{4}{9}\times\frac{3}{8}+\frac{1}{2}\times\frac{2}{5}\times\frac{1}{4}$

$=\frac{3}{36}+\frac{1}{20}$

$=\frac{1}{12}+\frac{1}{20}=\frac{1}{4}\left(\frac{1}{3}+\frac{1}{5}\right)$

$=\frac{1}{4}\left(\frac{8}{15}\right)=\frac{2}{15}$

(ii) $P(E)=\frac{\frac{1}{12}}{\frac{2}{15}}=\frac{1}{12}\times\frac{15}{2}$

$=\frac{5}{8}$

(iii) $P(E)=\frac{1}{2}\times\frac{5}{9}\times\frac{3}{5}+\frac{1}{2}\times\frac{3}{5}\times\frac{5}{9}$

$=\frac{1}{6}+\frac{1}{6}=\frac{2}{6}=\frac{1}{3}$

10.3 MATHEMATICAL EXPECTATION

If there are n possibilities $A_1, A_2, \ldots A_n$ in an experiment having the probabilities $p_1, p_2, \ldots\ldots p_n$ respectively. If value $M_1, M_2, \ldots\ldots, M_n$ are associated with the respective possibility. Then the ex pected value of the experiment is given by $\sum_{i=1}^{n} p_i \cdot M_i$

Q.1 A fair die is tossed. If 2, 3 or 5 occurs, the player wins that number of rupees, but if 1, 4, or 6 occurs, the player loses that number of rupees. Then find the possible payoffs for the player.

Ans.

Sol :

A_i	2	3	5	1	4	6
M_i	2	3	5	–1	–4	–6
P_i	1/6	1/6	1/6	1/6	1/6	1/6

Then expected value E of the game payoffs for the player

$$= 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) - 1\left(\frac{1}{6}\right) - 4\left(\frac{1}{6}\right) - 6\left(\frac{1}{6}\right) = -\left(\frac{1}{6}\right)$$

Since E is negative therefore game is unfavorable to the player.

Q.2 There are 100 tickets in a raffle (Lottery). There is 1 prize each of Rs. 1000/-, Rs. 500/- and Rs. 200/-. Remaining tickets are blank. Find the expected price of one such ticket.

Ans.

Solution : Expectation $= \sum p_i M_i$ outcome of a ticket can be

	p_i	M_i	$p_i M_i$
(i) I prize	$\frac{1}{100}$	1000	10
(ii) II prize	$\frac{1}{100}$	500	5
(iii) III prize	$\frac{1}{100}$	200	2
(iv) Blank	$\frac{97}{100}$	0	0
			$\sum p_i M_i = 17$

Hence expected price of one such ticket Rs. 17

10.4 GEOMETRICAL PROBABILITY

The following statements are axiomatic :

(a) If a point is taken at random on a given straight line AB, the chance that it falls on a particular segment PQ of the line is PQ/AB.

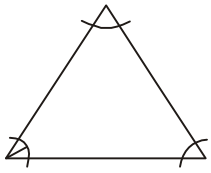
(b) If a point is taken at random on the area S which includes an area σ, the chance that the point falls on σ is σ/S.

Questions on Geometrical Probability

1. A point is selected at random inside an equilateral triangle whose length is 3. Find the probability that its distance to any corner is greater than 1.

Ans. $1 - \frac{2\pi}{9\sqrt{3}}$

Sol.



$$\text{Area} = \frac{\sqrt{3}}{4}\times 9 = \frac{9\sqrt{3}}{4}$$

$$\text{Area of sector} = \frac{1}{2}\times 1\times\frac{\pi}{3} = \frac{\pi}{6}$$

$$\text{Total area of sector} = 3\times\frac{\pi}{6} = \frac{\pi}{2}$$

$$\text{Favourable area} = \frac{9\sqrt{3}}{4} - \frac{\pi}{2}$$

$$\text{Probability} = \frac{\frac{9\sqrt{3}}{4} - \frac{\pi}{2}}{\frac{9\sqrt{3}}{4}} = 1 - \frac{\frac{\pi}{2}}{\frac{9\sqrt{3}}{4}}$$

$$= 1 - \frac{\pi}{2}\times\frac{4}{9\sqrt{3}}$$

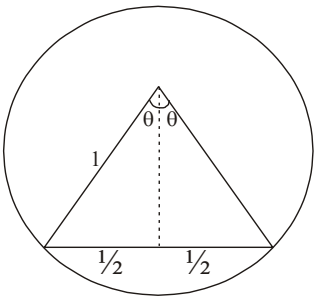
$$= 1 - \frac{2\pi}{9\sqrt{3}}$$

2. Two points are picked at random on the unit circle $x^2 + y^2 = 1$. The probability that the chord joining the two points has length at least 1, is

(A) 4/9 (B) 1/3 (C) 1/6 (D) 2/3

Ans. D

Sol.



$$\sin\theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}$$

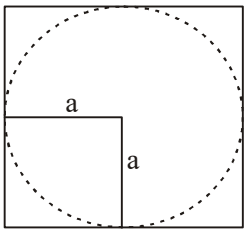
$$2\theta = \frac{\pi}{3}$$

$$\text{Probability} = \frac{\frac{2\pi}{3}}{\pi} = \frac{2}{3}$$

3. A circle of radius 'a' is inscribed in a square of side 2a. Find the probability that a point chosen at random is inside the square but outside the circle.

Ans. $1 - \frac{\pi}{4}$

Sol.



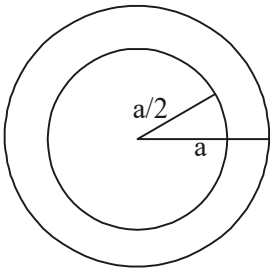
$$\text{Probability} = \frac{4a^2 - \pi \times a^2}{4a^2}$$

$$= 1 - \frac{\pi}{4}$$

4. A point is selected at random inside a circle. Find the probability that the point is closer to the circumference of the circle than to its centre.

Ans. $\frac{3}{4}$

Sol.



$$\text{Probability} = \frac{\pi a^2 - \pi\left(\frac{a}{2}\right)^2}{\pi a^2}$$

$$= 1 - \frac{\pi a^2}{4 \times \pi a^2}$$

$$= \frac{3}{4}$$

10.5 IMPORTANT CONCEPT

1. n whole numbers taken at random are multiplied together. Show that the chance that the digit at the unit place of their product is

(A) 1, 3, 7 or 9 is $\left(\frac{2}{5}\right)^n$ (B) 2,4,6 or 8 is $\frac{4^n - 2^n}{5^n}$

(C) 5 is $\frac{5^n - 4^n}{10^n}$ (D) 0 is $\frac{10^n - 8^n - 5^n + 4^n}{10^n}$

11.1 PROBABILITY DISTRIBUTION OF A RANDOM VARIABLE

A probability distribution spells out how a total probability of 1 is distributed over several values of a random variable

Q.1 A random variable X has the following probability distribution :

Determine

(i) k (ii) P(X < 3) (iii) P(X > 6) (iv) P(0 < X < 3)

Sol. **(i)** $P(0) + P(1) + P(2) + P(3) + P(4) + P(5) + P(6) + P(7) = 1$

$0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$

$10k^2 + 9k - 1 = 0$

$10k^2 + 10k - k - 1 = 0$

$k = \frac{1}{10}$ OR $k = -1$

(ii) $P(x < 3)$

$P(x = 0) + P(x = 1) + P(x = 2)$

$0 + k + 2k$

$= 3k$

$= \frac{3}{10}$

(iii) $P(X > 6)$

$P(X = 7) = 7k^2 + k$

$= \frac{7}{100} + \frac{1}{10} = \frac{17}{100}$

(iv) $P(0 < X < 3)$

$P(x = 1) + P(x = 2)$

$\Rightarrow k + 2k$

$\Rightarrow 3k$

$\Rightarrow \frac{3}{10}$

Q.2 A dice is thrown. X denotes number appeared on it. Find probability distribution of X.

Sol.

x	1	2	3	4	5	6
$P(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

Q.3 A coin is tossed 2 times. X denotes number of H appeared on it. Find probability distribution of X.

Sol.

x	0	1	2
$P(x)$	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{1}{4}$

Q.**4** 2 dice is thrown.X denotes sum of the numbers appeared on it. Find probability distribution of X.

Sol.

x	2	3	4	.....	12
$P(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	.....	$\frac{1}{36}$

Q.5 For binomial Probability distribution B(n,p), X denotes number of success. Find probability distribution of X.

Sol.

$x = r$	0	1	.....	n
$P(x)$	${}^{n}C_0P^0(1-p)^n$	${}^{n}C_1P^1(1-p)^{n-1}$	.....	${}^{n}C_nP^n$

11.2 MEAN AND VARIANCE OF A PROBABILITY DISTRIBUTION

Mean of any probability distribution of a random variable is given by :

$\mu = \frac{\Sigma p_i x_i}{\Sigma p_i} = \Sigma p_i x_i$ (Since $\Sigma p_i = 1$)

Variance of a random variable is given by,

$\sigma^2 = \Sigma (x_i - \mu)^2 \cdot p_i$

$\therefore \quad \sigma^2 = \Sigma p_i x_i^2 - \mu^2$ (Note that SD = $+\sqrt{\sigma^2}$)

Q.1 A coin is biased so that the head is 3 times as likely to occur as tail. This coin is tossed until a head or three tails occur. If X denotes the number of tosses of the coin, then the mean of X is :

एक सिक्का इस प्रकार अभिनत है कि चित्त के आने की संभावना पट् के आने की संभावना की तीन गुना है। इस सिक्के को तब तक उछाला जाता है जब तक कि एक चित्त या तीन पट् दिख जाएँ। यदि X सिक्के को उछालने की संख्या को दर्शाता है, तो X का माध्य है :

Ans. $\frac{21}{16}$

Sol. $P(H) = \frac{3}{4}$

$P(T) = \frac{1}{4}$

Mean $\overline{X} = \frac{3}{4} + \frac{3}{8} + 3\left(\frac{1}{64} + \frac{3}{64}\right)$

$= \frac{3}{4} + \frac{3}{8} + \frac{3}{16}$

$= 3\left(\frac{7}{16}\right) = \frac{21}{16}$

Q.2 Let X be a random variable such that the probability function of a distribution is given by $P(X = 0) = \frac{1}{2}, P(X = j) = \frac{1}{3^j} (J = 1, 2, 3, ..., \infty)$ Then the mean of the distribution and P(X is positive and even) respectively are :

माना X एक यादृच्छिक चर है जिसके लिए एक बंटन का प्रायिकता फलन $P(X = 0) = \frac{1}{2}, P(X = j) = \frac{1}{3^j} (J = 1, 2, 3, ..., \infty)$ द्वारा दिया गया है, तो बंटन का माध्य तथा P (X धनात्मक तथा सम है।) क्रमशः हैं :

Ans. $\frac{3}{4}$ and $\frac{1}{8}$

Sol. Mean = $\Sigma x_i p_i = \sum_{r=0}^{\infty} r \times \frac{1}{3^r} = \frac{3}{4}$

P (x is even) = $\frac{1}{3^2} + \frac{1}{3^4} + ...+\infty$

$= \frac{\frac{1}{9}}{1 - \frac{1}{9}} = \frac{\frac{1}{9}}{8/9} = \frac{1}{8}$

11.3 MEAN AND VARIANCE OF A BINOMIAL PROBABILITY DISTRIBUTION

The probability distribution for a binomial variate 'X' is given by :

$P(X = r) = {}^nC_r\, p^r\, (1-p)^{n-r}$ where P(X = r) is the probability of r successes.

Mean = np ;

variance = $np(1-p)$.

Q.1 A pair of dice is thrown 5 times. If getting a doublet is considered as a success, then find the mean and variance of successes.

Ans.

Sol : In a single throw of a pair of dice, probability of getting a doublet = $\frac{1}{6}$

considering it to be a success, $p = \frac{1}{6}$

$\therefore \quad q = 1 - \frac{1}{6} = \frac{5}{6}$

mean $= 5 \times \frac{1}{6} = \frac{5}{6}$, variance $= 5 \times \frac{1}{6} \cdot \frac{5}{6} = \frac{25}{36}$

Q.2 A pair of dice is thrown 4 times. If getting a total of 9 in a single throw is considered as a success then find the mean and variance of successes.

Ans.

Sol : p = probability of getting a total of 9 = $\frac{4}{36} = \frac{1}{9}$

$\therefore \quad q = 1 - \frac{1}{9} = \frac{8}{9}$

$\therefore \quad$ mean $= np = 4 \times \frac{1}{9} = \frac{4}{9}$

variance $= npq = 4 \times \frac{1}{9} \times \frac{8}{9} = \frac{32}{81}$

Q.3 Difference between mean and variance of a Binomial variate is '1' and difference between their squares is '11'. Find the probability of getting exactly three success

Ans.

Sol: Mean = np & variance = npq

therefore, $np - npq = 1$ .....(i)

$n^2p^2 - n^2p^2q^2 = 11$ .....(ii)

Also, we know that $p + q = 1$ .....(iii)

Divide equation (ii) by square of (i) and solve, we get, $q = \frac{5}{6}$, $p = \frac{1}{6}$ & $n = 36$

Hence probability of '3' success = ${}^{36}C_3 \times \left(\frac{1}{6}\right)^3 \times \left(\frac{5}{6}\right)^{33}$